

Model Equations

Objective

This document explains the regression equations used in the analysis, the meaning of each coefficient, the dummy variable approach, the reference category selected, and the final model chosen for business interpretation.

Dependent Variable

Monthly Sales

Monthly Sales is the dependent (response) variable because it represents the business outcome the retail chain wants to understand and improve.

Simple Regression Model 1

Independent Variable

Marketing Spend

Regression Equation

Monthly Sales = $\beta_0 + \beta_1(\text{Marketing Spend})$

Where:

- **β_0 (Intercept):** Expected monthly sales when marketing spend is zero.
- **β_1 :** Estimated change in monthly sales for every one-unit increase in marketing spend.

Business Interpretation

A positive coefficient indicates that higher marketing spending is associated with higher monthly sales.

A negative coefficient would indicate an inverse relationship.

Simple Regression Model 2

Independent Variable

Footfall

Regression Equation

$$\text{Monthly Sales} = \beta_0 + \beta_1(\text{Footfall})$$

Where:

- **β_0 (Intercept):** Expected monthly sales when footfall is zero.
- **β_1 :** Estimated increase in monthly sales for each additional customer visiting the store.

Business Interpretation

A positive coefficient suggests that increasing customer traffic is associated with increased sales.

Multiple Regression Model

Variables Included

Dependent Variable:

- Monthly Sales

Independent Variables:

- Marketing Spend
- Footfall
- Inventory Availability Percentage
- Customer Rating (if included)
- Region Dummy Variable (or Store Type Dummy Variable)

Regression Equation

$$\text{Monthly Sales} = \beta_0 + \beta_1(\text{Marketing Spend}) + \beta_2(\text{Footfall}) + \beta_3(\text{Inventory Availability}) + \beta_4(\text{Customer Rating}) + \beta_5(\text{Dummy Variable})$$

Replace the equation above with your actual regression equation after running the analysis.

Explanation of Each Coefficient

Intercept (β_0)

Represents the estimated monthly sales when all independent variables are equal to zero.

Although this situation may not occur in practice, the intercept serves as the baseline value for the regression equation.

Marketing Spend Coefficient (β_1)

Represents the expected change in monthly sales for each one-unit increase in marketing spending while holding all other variables constant.

Footfall Coefficient (β_2)

Represents the expected change in monthly sales associated with one additional customer visit while all other variables remain unchanged.

Inventory Availability Coefficient (β_3)

Measures the relationship between product availability and sales.

A positive coefficient indicates that better inventory availability is associated with higher monthly sales.

Customer Rating Coefficient (β_4)

Represents the estimated change in monthly sales associated with a one-point increase in customer rating while other variables remain constant.

Dummy Variable Coefficient (β_5)

Represents the average difference in monthly sales between the selected category and the reference category while holding all numerical variables constant.

Dummy Variable Approach

Categorical variables cannot be used directly in regression.

Dummy variables were created for one categorical variable.

Example:

Region

Suppose the dataset contains:

- North
- South
- East

- West

Three dummy variables are created:

- North
- South
- East

West is excluded.

West becomes the **reference category**.

The coefficient for each dummy variable measures the average sales difference relative to West.

Reference Category

Reference Category:

West Region (*Example*)

or

Standard Store Type (*if Store Type was used instead*)

The reference category is intentionally omitted to avoid perfect multicollinearity (the Dummy Variable Trap).

Why Dummy Variables Are Used

Dummy variables allow categorical information to be included in regression models while maintaining mathematical validity.

They help determine whether different regions or store types have systematically higher or lower monthly sales.

Final Model Selected

The **Multiple Regression Model** was selected as the final model because it:

- Includes multiple business drivers simultaneously.
- Provides stronger explanatory power (higher R^2).
- Controls for the influence of other variables.
- Produces more realistic business insights.
- Supports forecasting and strategic planning.

Business Interpretation

The regression equations estimate how changes in business variables are associated with changes in monthly sales.

Variables with statistically significant coefficients provide the strongest evidence of association and should receive the greatest attention during business planning.

However, regression identifies statistical relationships rather than proving cause-and-effect.

Business decisions should therefore combine regression findings with operational knowledge and additional validation.

Conclusion

The regression equations provide a structured method for estimating monthly sales based on important business factors. The multiple regression model is preferred because it captures several operational drivers simultaneously and produces more meaningful insights for retail decision-making while acknowledging that regression measures association rather than causation.